

Session 3.2

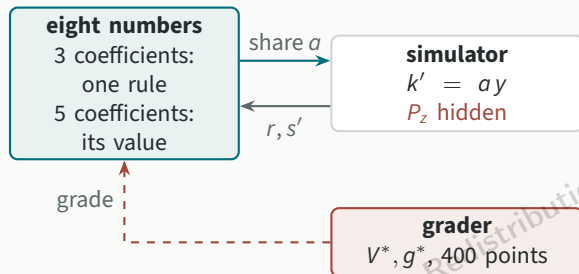
The Simplest Actor–Critic: How It Works, Why It Works, and the Wall It Meets

Nine numbers on the continuous growth model, with the shock probabilities hidden



Zhigang Feng

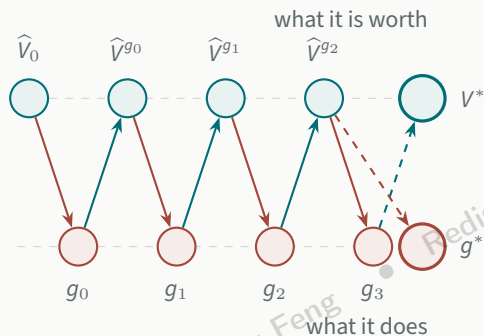
The Question This Deck Answers



- ▶ Capital is a real number and P_z is hidden. **Can eight numbers find the rule dynamic programming finds — on a 400-point grid, as the answer key?**
- ▶ **Hidden:** P_z . **Shown:** a realised reward and next state. **Kept by the grader:** V^*, g^* .
- ▶ No table, no grid, no network: **one saving rule with three coefficients, and one value function with five.**

Source: The model of Deck 3.1, frame 3; $\alpha = 0.36, \beta = 0.95, \delta = 0.10, u = \ln c, z \in \{0.9, 1.1\}$ with switching probability 0.2. Answer key and learner: Session 3 lab, block A2.

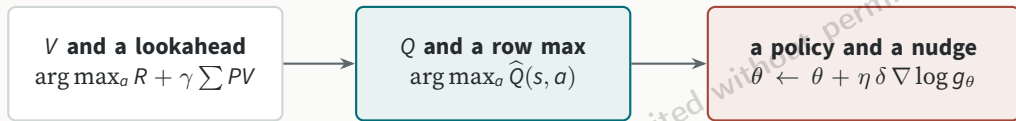
Evaluate, Then Improve: the Loop Every Method Here Runs



- **Generalized policy iteration:** alternate *evaluation* (teal) and *improvement* (red).
- **Howard's policy iteration** evaluates a rule *to convergence* before improving it; value iteration takes one backup instead.
- **This deck takes one *sampled* backup and one nudge, every transition — three points on one dial.**

Source: R. Howard, *Dynamic Programming and Markov Processes*, 1960; Sutton & Barto, 2018, §4.6; Miao & Feng, ch. 11.

Improving a Rule Means Comparing Actions — and a Third Road That Does Not



needs P

dynamic programming

needs a max

SARSA, Q-learning

needs neither

actor-critic

- ▶ **The left two are the 7×2 grid's.** With V you must look one step ahead, which needs P . With Q the comparison is already inside the table: **read the row for the state you are in, take its largest entry** — no model, no derivative, just a comparison.
- ▶ **The right one is this deck's.** On the continuous model k is real, so there is no cell to hold Q ; the share is real, so there is no finite row to scan.
- ▶ A max is still available over a fine grid of shares, as the key takes — but only with the model, or a learned $\hat{Q}$ (Deck 3.5).
- ▶ **A gradient needs neither, and it survives Session 4.**

Source: Miao & Feng, ch. 12, “From value-based to policy-based methods”. Deck 3.1 draws the row.

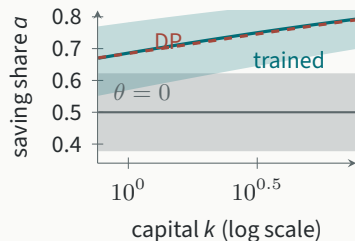
Session 3.2

The Simplest Actor-Critic

Four numbers choose, five numbers judge, one number moves both



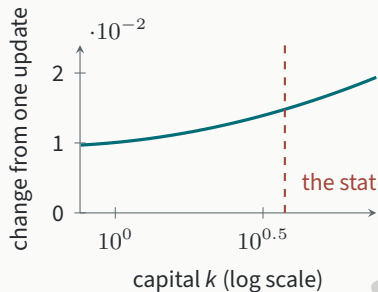
The Actor: a Saving Rule, Three Numbers



- ▶ **The rule is a deterministic function of the state**, as the growth model writes one:
 $a = \text{sigmoid}(\theta_0 + \theta_1 k + \theta_2 \mathbf{1}\{z = 1.1\})$. **Three numbers.**
- ▶ With no model you cannot learn what a *different* share is worth without trying one, so while training we **probe**: draw around the rule at a fixed width — the bands. **The probe is machinery, not the rule**: not learned, switched off to grade.
- ▶ It starts flat at 0.5000 and ends on DP's curve. **The band never narrows; what moves is the line.**

Source: Session 3 lab, block A2; $z = 0.9$, untrained and at 10^6 steps. Bands: one probe width either side.

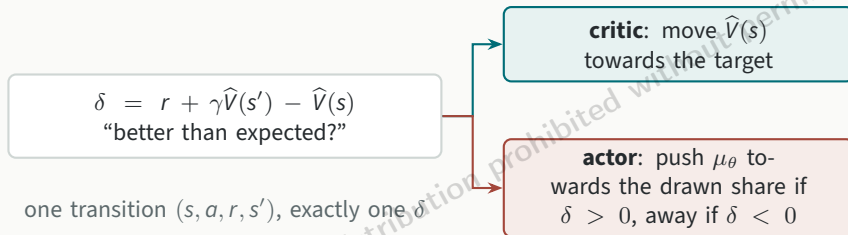
The Critic: Five Numbers, and What One Update Does



- ▶ $\hat{V}_\psi = \psi^\top \phi$, with $\phi = (1, \tilde{k}, \tilde{k}^2, \mathbf{1}\{z = 1.1\}, \tilde{k} \mathbf{1}\{z = 1.1\})$.
- ▶ It starts flat at $u(c^*)/(1 - \gamma) = 4.27$, while the true V^* runs from -0.25 to 6.48 across the band.
- ▶ The plot is what *one* update did. The transition happened at $k = 3.75$, and **the value moved at every other k too** — 0.010 at the bottom of the band, 0.019 at the top. **There is no cell to correct.**
- ▶ The target $r + \gamma \hat{V}_\psi(s')$ uses the critic's own estimate: **bootstrapping**.

Source: Session 3 lab, block A2; the first update of the worked period on frame 8.

The TD Error: One Number, Two Jobs



- ▶ δ compares what happened with what the critic expected. It estimates the **advantage** $Q^g(s, a) - V^g(s)$: a grade for the action.
- ▶ Session 2 warned that sampling an *inner* expectation is not harmless — it sat inside a logarithm and a square. **Here the update is linear in the target, so a draw is unbiased.**
- ▶ **Evaluation and improvement from one number — frame 2's loop at full speed.**

Source: Miao & Feng, *AI for Economists*, ch. 12, "Actor-critic"; Sutton & Barto, 2018, §13.5.

The Whole Algorithm on One Page

One-step actor-critic, in the book's notation

- 1 initialise θ (three numbers), ψ (five); fix the probe width σ ; draw s from the band
 - 2 **for** $t = 0, 1, 2, \dots$:
 - 3 probe the rule: $x \sim N(\mu_\theta(s), \sigma^2)$, $a = \text{sigmoid}(x)$; the simulator returns r and s'
 - 4 $\delta \leftarrow r + \gamma \hat{V}_\psi(s') - \hat{V}_\psi(s)$ *at a reset, bootstrap as usual*
 - 5 $\psi \leftarrow \psi + \eta_\psi \delta \phi(s)$ *critic: one step*
 - 6 $\theta \leftarrow \theta + \eta_\theta \delta \nabla_\theta \ln g_\theta(a | s)$ *actor: one step*
 - 7 every 300 periods redraw s from the band; otherwise $s \leftarrow s'$
- The **score** closes in one expression, $\nabla_\theta \ln g_\theta(a | s) = \frac{x - \mu}{\sigma^2} (1, \tilde{k}, \mathbf{1}\{z = 1.1\})$. The sigmoid's Jacobian does not depend on θ , and σ is not a parameter, so nothing else appears.
 - **Line 4 is the only place the future enters, and it enters through the critic.**

Source: Miao & Feng, ch. 12, §12.5.1, “one-step actor-critic”; the comment on line 4 is the book's.

A Worked Period: One Transition, Every Number

```
state      k = 3.7500  z = 0.90      ktilde = +0.3814
actor      mu = 0.0000  sigma = 0.5000  eps = +0.1257  ->  x = +0.0629
action     a = sigmoid(x) = 0.5157      (DP's share here: 0.7549)
model      y = 4.8234  k' = a y = 2.4875  c = (1-a) y = 2.3359
reward     r = ln c = +0.8484            next shock z' = 0.90
delta      r + gamma V(s') - V(s) = +0.6347  (V(s) = V(s')) = 4.2732
```

```
psi  4.2732 0 0 0 0  ->  4.2859  0.0048  0.0018  0  0
theta 0 0 0          ->  0.000032  0.000012  0      (probe fixed at 0.5000)
```

- ▶ The rule drew 0.516 where DP saves 0.755, so it under-saved; the reward was still good, the critic had expected less, and $\delta > 0$ pushed μ_θ towards what was drawn.
- ▶ **The critic's constant moved 0.0127; the actor's moved 0.000032 — four hundred times slower.** That is the two-time-scale design: the critic must know what a state is worth before the actor is allowed to believe it.

Source: Written by the lab so the slide cannot drift, seed 0, untrained start; Session 3 lab, block A2.

Why No Transition Matrix Appears Anywhere

a path's probability $\Pr(\tau) = \underbrace{\mu(s_0) \prod_t P(s_{t+1} \mid s_t, a_t)}_{\text{the environment}} \cdot \underbrace{\prod_t g_\theta(a_t \mid s_t)}_{\text{the rule}}$

the environment's terms do not contain θ

take logs, differentiate in θ $\nabla_\theta \log \Pr(\tau) = \sum_t \nabla_\theta \log g_\theta(a_t \mid s_t)$

- ▶ P enters the probability of a path but not its *derivative*: it differentiates away.
- ▶ The critic needs no P either — its target $r + \gamma \hat{V}(s')$ is built from one realised draw.
- ▶ **This is why the method survives challenge 2. It is also Deck 3.5's policy-gradient theorem, one slide early.**

Source: Miao & Feng, *AI for Economists*, ch. 12, “The policy gradient theorem”; Sutton & Barto, 2018, §13.2.

Session 3.2

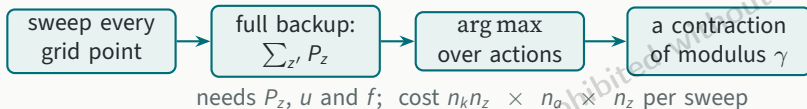
Why It Works, and What Survives

The contraction is still there — for a fixed rule

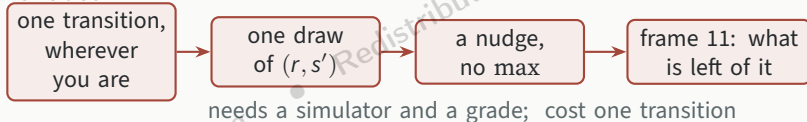


Compared with Dynamic Programming, Drawn

dynamic programming



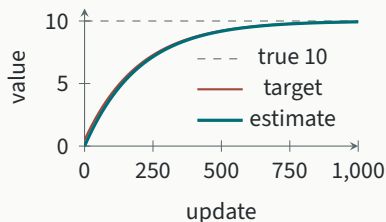
this deck



- ▶ Every column is a trade: the sweep for a visit, the sum for a draw, the maximum for a nudge.
- ▶ **The first three make challenge 2 passable. The fourth is the bill.**

Source: Miao & Feng, ch. 11 and ch. 12, §12.6.1.

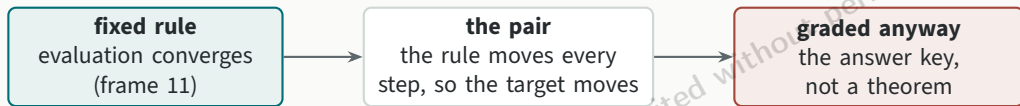
What Survives: the Contraction, for a Fixed Rule



- ▶ Hold the rule fixed. The expected target is $T^g \hat{V}$ — Deck 3.1's operator, a contraction of modulus γ , and **sampling applies it on average**.
- ▶ With *this* critic it reaches the **projected fixed point**: the best linear fit to $T^g \hat{V}$ over the states the rule visits, within $1/(1 - \gamma) = 20$ of that fit — or ≈ 3.2 by the book's tighter argument.
- ▶ The binding condition is that the data come from the rule being evaluated. Ours do, up to a reset every 300 periods, and $\gamma^{300} \approx 2 \times 10^{-7}$.
- ▶ **The max threw the contraction away. A fixed rule gets it back.**

Source: Tsitsiklis 1994 and Jaakkola, Jordan & Singh 1994 (tabular); Tsitsiklis & Van Roy 1997 (linear) — the book's theorems 12.1 and 12.2.

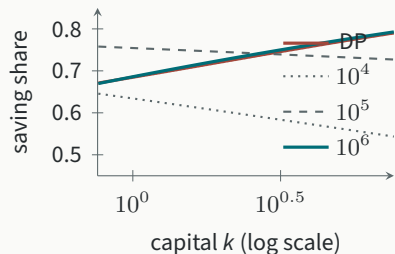
What Does Not Survive, and What Replaces It



- ▶ Once the actor moves, the critic chases a moving target and Howard's finite termination is gone.
- ▶ What is known covers neighbouring cases — exact gradients on softmax rules, two time scales with a linear critic — **under conditions this run does not meet**, which is what the book says of its own chapter's algorithms.
- ▶ **So the guarantee this session runs on is the answer key.** Deck 3.5 frame 18 states the conditions and closes the debt.

Source: Konda & Tsitsiklis, *NeurIPS* 12, 2000; Agarwal, Kakade, Lee & Mahajan, *JMLR* 22, 2021; Mei et al., *ICML*, 2020. Miao & Feng, ch. 12, §12.5.1 footnote.

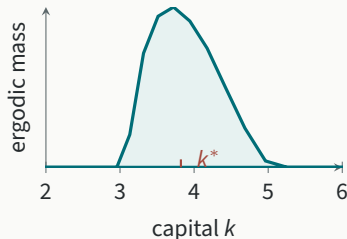
Measured: the Rule It Learned, against the Rule DP Found



- ▶ $z = 0.9$; the $z = 1.1$ panel is the same picture — the learned shift is -0.001 .
- ▶ By 10^6 steps the rule sits on DP's curve, and **it finds the slope, not just the level:** $\theta_1 = 0.317$ against the 0.320 that is the best this shape can do.
- ▶ **That best rule scores** 0.00001% , so everything on the next frame is *learning* error.
- ▶ **The critic is not accurate** — rms 1.72 against a V^* running -0.25 to 6.48 . It does not need to be: **it only has to rank actions.**

Source: Session 3 lab, block A2; median of 20 seeds. Continuous model.

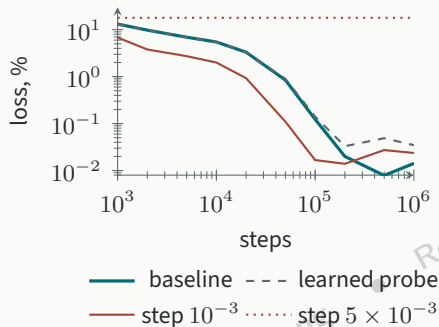
Where the Economy Lives, and the Grade



- ▶ A rule is graded where its own economy goes: here between $k = 3.0$ and 4.8 , though the band runs to 7.6 .
- ▶ **Median of 20 seeds: 0.014 % where the economy lives, 0.015 % over the band; best seed 0.002 %, worst 0.106 %.**
- ▶ A **consumption-equivalent welfare loss**: the fraction of consumption given up, for ever, to hold the learned rule instead of DP's.
- ▶ **Session 2 used those words for Judd's Euler-residual scale — a different object.**

Source: Session 3 lab, block A2; the probe is off when the rule is graded, and the density comes from Young's two-point split.

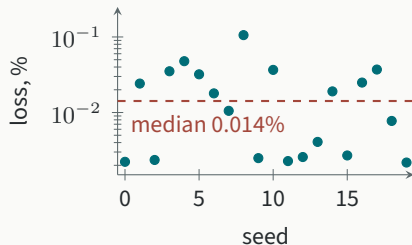
Three Knobs Decided the Answer



- ▶ **Actor step.** 2×10^{-4} gives 0.014 %; 10^{-3} , 0.024 %; 5×10^{-3} destroys the run.
- ▶ **Probe width.** 0.5 is tuned: at 0.25 the rule finds the level but not the slope ($\theta_1 = 0.17$ against 0.32); at 1.0 the run is destroyed.
- ▶ **Learning the width instead costs 0.014 %** → **0.035 %** — σ falls to 0.13 and the rule stops trying alternatives. **That is why the probe is a knob, not a parameter.**
- ▶ **Resets.** 0.014 % with, 0.018 % without — *exploring starts by another name.*

Source: Session 3 lab, block A2; medians of 20 seeds at 10^6 steps.

When It Fails, How It Fails



- ▶ Twenty seeds, same settings: 0.002 % to 0.106 %. **One run is not a measurement.**
- ▶ **Every one of the twenty over-saves**, and the spread is the *level*, not the shape: the error in θ_0 tracks the loss almost exactly, the error in θ_1 not at all.
- ▶ Under- and over-saving of the same size cost the same, so **the bias is in the estimator, not the objective.**
- ▶ The first place to look is the γ^t weight the theorem asks for and this demo drops — **Deck 3.5 puts a price on exactly that.**

Source: Session 3 lab, block A2; 20 seeds, 10^6 steps. Weighted share error against DP's rule.

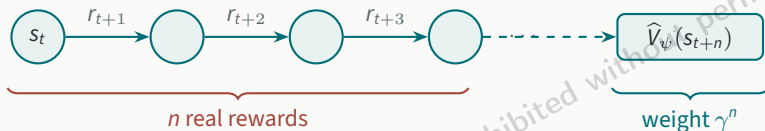
Session 3.2

Improve It: Roll Out T Periods

And the wall that puts in the way — uncertainty, unfolded



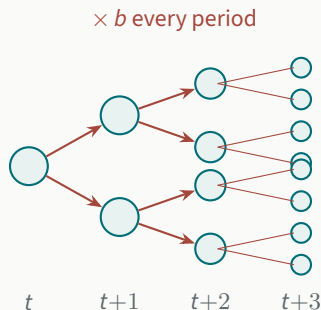
Why Roll Out at All



- ▶ Both halves of the update lean on the critic. With $n = 1$ it carries weight $\gamma = 0.95$; take n real rewards first and its weight falls to γ^n .
- ▶ The **n -step return** is $\hat{G}_{t:t+n} = \sum_{j < n} \gamma^j r_{t+j+1} + \gamma^n \hat{V}_\psi(s_{t+n})$, and the advantage becomes $\hat{G}_{t:t+n} - \hat{V}_\psi(s_t)$.
- ▶ In dynamic programming this is **modified policy iteration**. It is not Howard's improvement theorem.
- ▶ **Fewer numbers taken on trust. The next two frames are what it costs.**

Source: Miao & Feng, ch. 11, §11.4.2, eq. (11.23) and its footnote. The usual attribution for modified policy iteration is Puterman & Shin, *Management Science* 24, 1978.

The Unfolding of Uncertainty

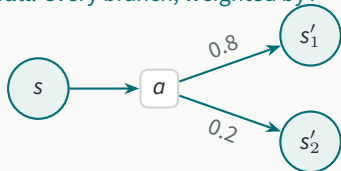


- ▶ On a table the rollout is a recursion: histories re-merge on the grid. **The key's grid re-merges; the learner's path does not.**
- ▶ From one state, T periods ahead is a tree of b^T shock histories: $2^{20} \approx 10^6$ here, $7^{20} \approx 8 \times 10^{16}$ with seven shock nodes.
- ▶ **One more state variable multiplies the table by m ; one more period multiplies the tree by the branching factor b .**
- ▶ **A second curse, in time — created by the answer to challenge 2.**

Source: Miao & Feng, ch. 11, §11.4.2.

Sampling the Tree

full: every branch, weighted by P



sample: one branch, no probabilities

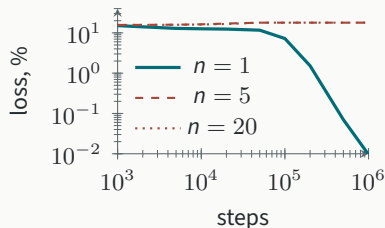


Source: Miao & Feng, ch. 11, §11.4.3.

	full	sample
reads	P	realised (r, s')
T steps ahead	b^T	$N \times T$
error	none	$\sigma_G / \sqrt{N}$
grows with d, b ?	yes	no

- ▶ A **sample backup** replaces the sum by walks: N of length T , cost $N \cdot T$, error $\sigma_G / \sqrt{N}$ — free of both b and d .
- ▶ Large T is **Monte Carlo** (3.3); $T = 1$ is **temporal difference** (3.4).
- ▶ **The rest of the session is the choice of T , and of what to store.**

Measured: Rolling Out Made It Worse



- ▶ Four actor steps tried per n , each curve that n 's best. **Only $n = 1$ learns**; $n = 5$ and $n = 20$ end at the band's floor at every step tried.
- ▶ Not the code — with the actor fixed, n -step evaluation reaches the same critic for every n ; nor the critic's step, which can move tenfold and change nothing.
- ▶ **It is the rule's own path**: with $n = 20$, twenty periods of the *current* rule run before the critic is consulted, and early on that rule wastes capital.
- ▶ **Not bias against variance, but the bias a bad rule puts into its own long return.**

Source: Session 3 lab, block A2; medians of 20 seeds. A collapsed run grades 17.9 %: the rule pinned at the band's floor.

Session 3.2

Why This Is Not Yet Enough

What eight numbers bought, and what they did not



The Other Road, on a Grid



- ▶ On the 7×2 grid the maximum is cheap and it is the better road: **Deck 3.4 introduces SARSA and Q-learning**, and Deck 3.6 puts them on the same grade as this deck.
- ▶ This deck could not take that road, because a table needs cells and this model has none — which is the honest reason, not a preference.
- ▶ **The two roads swap places in Session 4.** There the action is a whole saving function; there is no row, and no grid over it either.

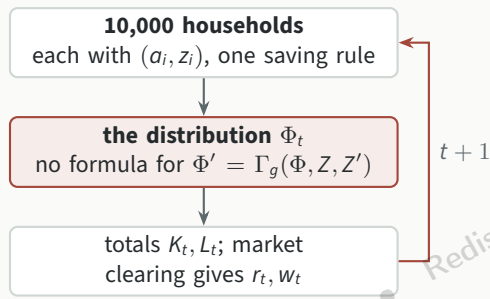
Source: Miao & Feng, *AI for Economists*, ch. 12, §§12.2 and 12.4.

What the Actor–Critic Bought, and What It Did Not

against dynamic programming	where it is paid for
✓ reads samples, never P_z	challenge 2, frame 9
✓ no maximum over actions	frame 3
✓ no grid over the state — eight numbers	frame 5
× the rule and the value are only as good as their <i>shape</i>	Deck 3.5
× a rollout unfolds uncertainty	frames 18–20, Decks 3.3–3.4
× guarantees: evaluation yes, improvement conditional	frame 12, Deck 3.5

- ▶ Frame 10 drew the mechanics. **This is the bill.**
- ▶ **Three ticks, three crosses, and the crosses are the rest of the session.**

The Model Economists Actually Want



- ▶ Krusell–Smith: the state becomes $(a_i, z_i; Z, \Phi)$, and Φ as a histogram is n_b numbers, not two.
- ▶ **A rule linear in five features cannot carry that** — the features must be learned (3.5).
- ▶ Φ 's law of motion must come from somewhere: the KFE in continuous time, an analytic differentiable kernel in discrete time, or a simulated panel. **Deck 3.6 builds one.**
- ▶ **Nine numbers were never the point. The loop was.**

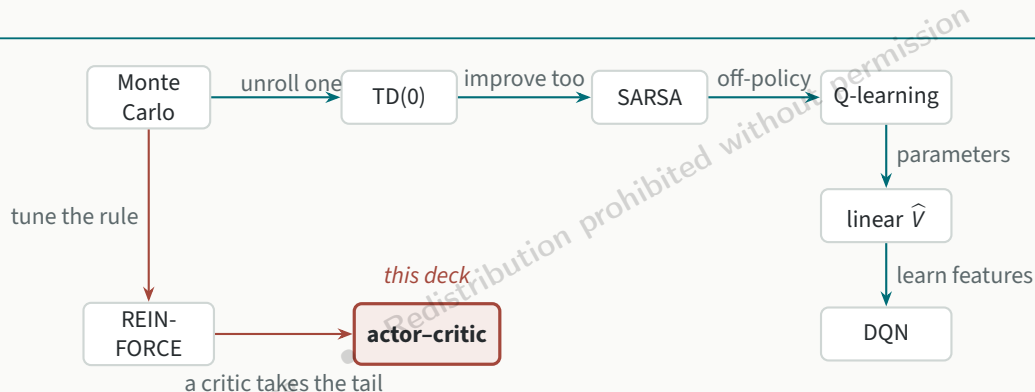
Source: Krusell & Smith, *JPE* 106(5), 1998; Miao & Feng, ch. 12, §12.7.

Nine Numbers, Six Ideas, and Where Each Is Earned

in the demo	the idea, used on credit	earned in
one draw of (r, s') per update	sample the expectation	3.3, frames 1–3
the target $r + \gamma \hat{V}_\psi(s')$	bootstrap one step; how far to roll out	3.4, frames 3–6
$\hat{V}_\psi = \psi^\top \phi$, five numbers	approximate the value: features, then networks	3.5, route one and two
a saving rule, three numbers	improve by a gradient, not a max	3.5, policy gradient
both moved by one δ	evaluation inside improvement	3.5, actor–critic
a fine-grid DP as the grade	an answer key before any learner runs	3.1, then 3.6

- ▶ **Every row is something this deck used without earning.** Each of the next four decks opens by naming the row it pays for — and **Deck 3.6 closes this table.**

The Session's Map, with This Deck Lit



- ▶ Axes: value-centred → policy-centred, tables → parameters. **Every arrow is one deck of this session, and its label is what that step changed.**
- ▶ **This deck sits at the far corner, and got there in one jump.**

Source: After Miao & Feng, ch. 12, figure 12.1; the arrow labels are the book's.